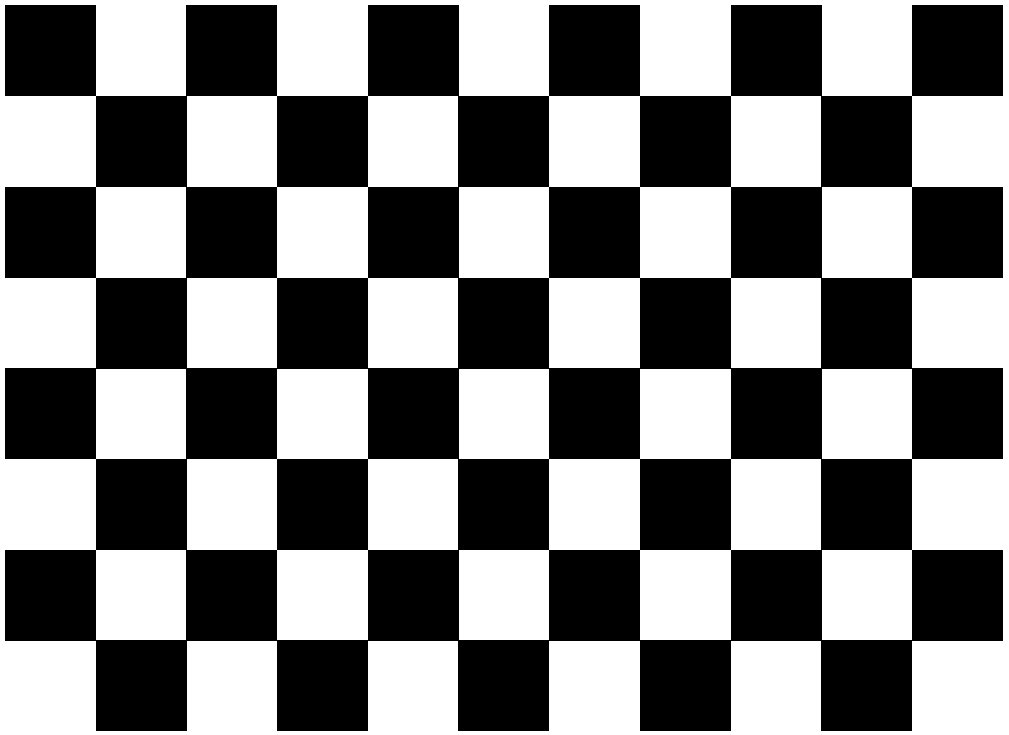

V: must be 120.0 mm

H: must be 120.0 mm

SCALE CHECK - measure BOTH rulers before use: H (horizontal) AND V (vertical) must EACH measure 120.0 mm (10 squares).

If either does not, do NOT use this board: fix the printer scaling (100 %, no fit-to-page) and reprint. Why both: page 2.

11 x 8 squares, 12 mm nominal, 10 x 7 inner corners. Solve: python Example/Camera/calibrate.py --images <DIR> --board-squares 11x8 --square-mm <CALIPERED mm> --board-height-mm <PLATE + PAPER mm>

Checkerboard calibration target - 11 x 8 squares, 12 mm

1. Print

Print page 1 at 100 % (actual size): no "fit to page", no "shrink to printable area", landscape, on matte paper. Glossy paper glares under the flat light and hides corners.

2. Check the scale - both axes, before anything else

With a steel rule, measure the H and V rulers on page 1. EACH must be 120.0 mm (10 squares). If either is not, do NOT use the board: fix the printer scaling and reprint.

Both, not one: a printer can scale the paper-feed direction differently from the head direction, and calibrate.py solves the x and y mm/px factors separately from ONE square size. An error in one direction goes straight into that axis's factor, and nothing in the calibration report will show it.

Then caliper the board across 10 squares horizontally AND vertically; --square-mm is that length / 10. If the two directions differ by more than 0.1 mm (0.083 %, under the calibration's 0.1 % uniformity limit), reprint: one --square-mm cannot carry two scales.

3. Mount - the flatness requirement grows with the board

Mount the 132 x 96 mm board on a rigid plate: glass, or aluminium at least 3 mm thick. Card and foam board bow, and a larger print needs a stiffer plate still. A warped board produces a calibration that looks fine and is WRONG: the solver assumes a plane, the bow can be absorbed into the lens model at a small reprojection error, and every measurement inherits it.

Glue the whole sheet down without bubbles. Keep the white border around the board intact and unmarked: the detector needs about one square of white on every side. Do not trim it, write on it or tape over it.

4. Calibrate

Capture at least 40 views, flat (f) and tilted (t); the capture checklist, item 1, says how many of each and why 40:

```
python Example/Camera/calibrate.py --capture --board-squares 11x8
```

Solve:

```
python Example/Camera/calibrate.py --images <DIR> --board-squares 11x8 --square-mm <CALIPERED mm> --board-height-mm <PLATE + PAPER mm>
```

<CALIPERED mm> is the square size from step 2 - never the nominal 12 mm printed here. <PLATE + PAPER mm> is the board surface's height above the table: the plate and paper thickness, measured.

The board

11 x 8 squares (--board-squares 11x8), 10 x 7 inner corners, 12 mm nominal squares, pattern 132 x 96 mm. Generated by Example/Camera/generate_checkerboard.py.